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Rehabilitation and Prosthetic Restoration in Upper Limb Amputation

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SUMMARY PEARLS

- Prognosis and adaptation: Upper extremity amputees generally have a good prognosis for functional restoration, especially with modern rehabilitation and prosthetic technology.
- Terminology: Preferred terms include *residual limb* rather than *stump* and *limb loss* or *limb difference* over *amputation*. Terminology should be patient specific.
- Classification of amputation: Upper limb amputations are classified into digit amputations (most common), minor amputations (below the wrist), and major amputations (at or above the wrist).
- Prosthetic options: Four categories of upper limb prostheses exist: passive, body powered, myoelectric, and hybrid. Each is tailored to the patient's functional and vocational goals.
- Prosthetic training phases: Prosthetic training is divided into preprosthetic, early prosthetic, and advanced prosthetic phases; each is aimed at maximizing functional adaptation.
- Pain management: Residual limb pain and phantom limb pain must be addressed during rehabilitation to prevent prosthetic abandonment.
- Financial considerations: The cost of prosthetics can range from \$3,000 to over \$40,000, and insurance coverage varies. Patients should be guided through the financial realities of prosthetic care.
- Special considerations: Bilateral amputees, children, and patients with additional impairments such as traumatic brain injury may require specialized prosthetic devices and training approaches.

Introduction and Preferred Terminology

The loss or absence of a limb is a unique and profound condition in which a person has lost a very tangible part of oneself. Many patients and even health care providers have an understandable anxiety about the future following amputation. In fact, when patients with diabetes were asked about what complications they fear most, they rated amputation as more frightening than death.⁴⁵ With today's rehabilitation and prosthetic technology, this sobering outlook need not be the case, especially for individuals with upper extremity limb loss, who in many cases have a good prognosis for functional restoration and adaptation. In many cases, amputation can provide the opportunity for better function. Even patients with bilateral amputation or significant comorbidities can lead fulfilling and active lives. Despite the high visibility of the condition, amputees are frequently underserved. Even within health care systems with well-developed amputee rehabilitation programs, patients with limb loss may never receive specialized amputation rehabilitation care or even receive prosthetic training.³⁶ We hope that this chapter aids our readers in meeting this unmet need by improving care within your sphere of practice for patients with limb loss and limb difference.

This chapter provides a comprehensive overview of the evaluation and management of upper limb loss and limb difference patients, with a focus on relevant information to physiatrists and

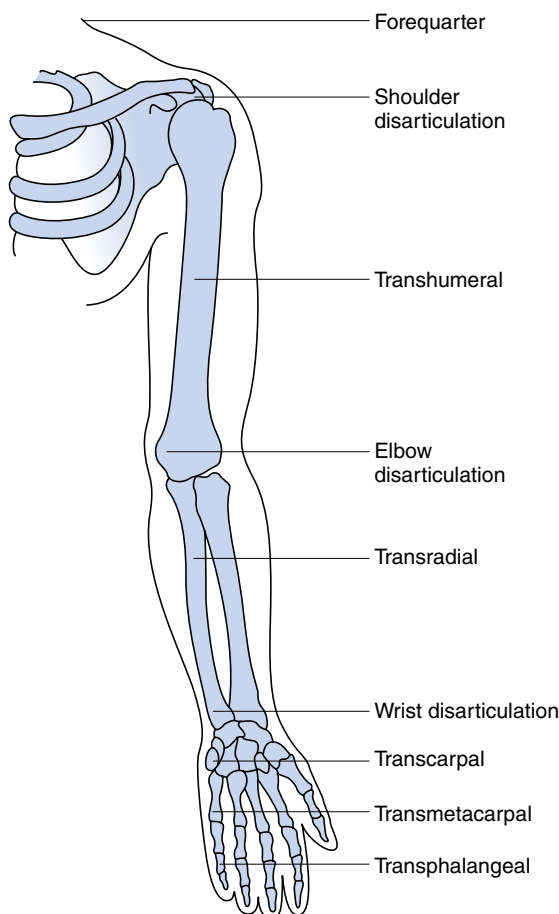
other rehabilitation professionals. We will discuss background information regarding both adult and pediatric amputation and limb differences, surgical considerations and practices, prosthetic evaluation and prescription, as well as the management of comorbidities and rehabilitation complications as the patient progresses from the surgical stage to acute rehabilitation and then to long-term follow-up. We will discuss the impact of emerging technologies on the field and important considerations for reintegration and adaptation to limb loss.

Terminology in clinical practice, such as *amputation* and *limb loss*, are used interchangeably in this chapter, as well as the equivalent terms *amputee* and *person with limb loss*. The term *residual limb* or *residuum* is generally preferred as the medical term for the post-amputation limb, replacing older common terms such as *stump*. The *sound limb* conventionally refers to the nonamputated side. The term *limb difference* denotes a patient with a nonstandard limb who did not undergo an amputation, but instead has a congenitally absent or malformed limb. Terminology preference is patient and context specific and evolves over time. What matters most is that health care providers and patients understand one another.

Limb Loss Classification

The anatomy of various amputation locations is an important first step in understanding their functional impact and rehabilitation

strategies. The most common types of upper extremity amputations are depicted in Fig. 10.1. Digit amputations are the most common (78%) of upper extremity amputations.¹⁰ By convention, a *minor* limb amputation is one occurring below the wrist, while those occurring at or above the wrist are termed *major* limb amputation. A digit amputation's functional effects depend on the digit involved and the activities of the patient. The thumb is critical for palmar grip, adduction pinch, and oppositional pinch, but it may be replaced with a thumb driver or similar prosthesis to restore function. The index finger interacts intimately with the thumb for many of these functions, but if lost or greatly shortened, this function can often be subsumed by the third digit. Amputation involving the third and fourth digits is less functionally impactful and is often restored with a primarily cosmetic prosthesis. The fifth digit's importance and strength in hook grasp is often underappreciated, especially for patients participating in manual labor. From a surgical perspective, retaining a greater length of any of these digits will improve function. In some cases, however, a hand may be more functional with a more extensive ray amputation than a very short residual finger.³⁰ Surgical reconstructive techniques, such as phalangization or pollicization, can rearrange and modify remaining digits to maximize function in some patients. Wrist disarticulations provide a greater degree of supination and pronation than more proximal transradial amputations, but they present prosthetic difficulties due to their long length, making prosthesis use difficult and arm length asymmetric to the sound limb.



• Fig. 10.1 Types of upper extremity amputations.

Transradial amputations are the most common type of major upper limb amputation and are subcategorized by their residual bone length from the end of the limb to the medial epicondyle. Generally, a longer limb retains more supination/pronation, about 50% of which can be transferred to a prosthesis. A longer limb also provides a longer lever arm for elbow flexion strength.

Elbow disarticulations are less common and are most frequently seen in pediatric patients to preserve the bone's terminal growth plate. While they can be accommodated by modern techniques for cosmesis, the residual limb is difficult to fit with a natural and functional prosthetic elbow joint due to its length and may result in arm asymmetry.

Transhumeral amputations are also classified by their length, with a longer limb preserving more glenohumeral motion and strength. As more of the limb's length is reduced, the prosthetic socket must encompass more of the shoulder and inhibit range of motion. At levels of amputation higher than the humeral neck, shoulder range of motion is largely lost. This includes glenohumeral disarticulation and more proximal forequarter amputation, which are normally performed for local malignancy or severe trauma. While these amputations present difficulties in fitting a prosthesis, newer myoelectric technology is broadening the functional possibilities for these patients, as we will discuss later in the chapter.

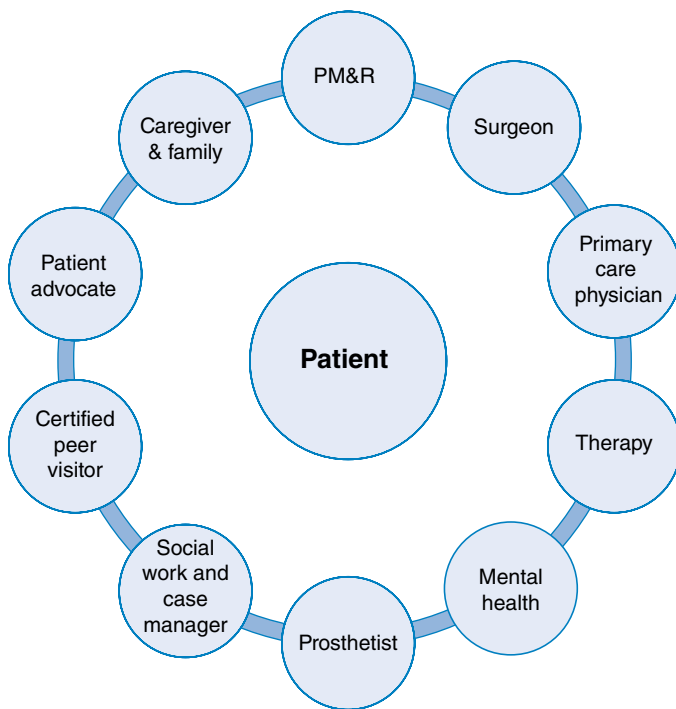
Demographics, Incidence, and Prevalence of Adult Limb Loss

About 185,000 persons undergo an amputation in the United States yearly. In 2008, the prevalence of all limb loss nationwide was estimated at 1.9 million people, of which about 500,000 are living with a minor upper limb amputation and 41,000 with a major upper limb amputation.⁴ While the number of people living with lower extremity limb loss in the United States is expected to double by the year 2050 (largely due to rates of dysvascular disease, obesity, and diabetes), the etiologies of upper extremity limb loss differ significantly, and their incidence is generally expected to remain stable or slightly decrease over the coming years, partly thanks to improved workplace safety standards and advancements in surgical limb salvage, limb replantation, and even limb transplantation. Of upper limb amputations, 80% are due to trauma, with most occurring at work involving saws, blades, or other machinery in the civilian population. Active military personnel experience a higher rate of amputations due to blast or gunshot injuries and have a higher incidence of concomitant traumatic brain injury, multiple limb amputation, visual and hearing loss, soft tissue loss, and burns.⁹

Other causes of upper limb loss include burns, peripheral vascular disease and clot-related ischemia, infection, and malignancy. There are also rare cases of planned, elective amputation in unusual scenarios such as severe neurologic injury causing limb paralysis, severe nonfunctional contracture, and congenital deformity. In general, males account for more than 75% of upper limb loss, and the more severe the injury the more likely the victim is male.²

The Rehabilitation Team and the Role of Physical Medicine and Rehabilitation

As with other types of rehabilitation patients, those with limb loss are most successful when cared for by a well-coordinated interdisciplinary team of professionals (Fig. 10.2). The surgical team varies depending on the etiology of amputation, but may include



• **Fig. 10.2** The limb loss interdisciplinary rehabilitation team.

trauma, orthopedics, plastic, and vascular surgeons. The primary care physician is an indispensable part of the team for managing general medical care and comorbidities and helping to coordinate referrals. The physical and occupational therapists ideally have experience working with upper extremity amputees, with or without a prosthetic device. These professionals help ensure a patient is ready to manage the activities of daily life and help to recommend assistive equipment. Psychiatry and psychology professionals can offer counseling and support services for the patient and families and can help process the emotional journey of limb loss and limb difference, and help to connect the patient to community and vocational resources.

The prosthetist is a particularly important part of the team. The prosthetist evaluates the patient and then recommends, fabricates, and fits the prosthesis for the patient. Going forward, the prosthetist offers important follow-up care for adjustments and troubleshooting of prosthetic issues and acts as a resource to the health care provider and patient to help navigate the expanding variety of prosthetic componentry available today.

Other team members in the community include patient advocates, social workers and case managers, and certified peer visitors. When available, a professional patient advocate can be helpful to a patient in navigating the health care system, particularly around issues of insurance coverage and appointment access. Similarly, a social worker or case manager can assist in several ways, such as coordinating insurance communication, preparing follow-up appointments, and ordering equipment after discharge. In cases where workers' compensation is involved, this case manager may also help coordinate coverage and compensation for the patient.

A certified peer visitor (CPV) is an individual outside of the patient's family or care team that has usually had a similar experience and can provide support and guidance but not medical advice. The CPV may have similar limb loss or limb difference and may have experience as a caregiver. Community organizations,

such as the Amputee Coalition of America (ACA), can offer additional resources and connect to local support groups with community and recreational opportunities. The ACA also administers the CPV training program, and a CPV can be requested by the patient or care team through the website. Family members and other caregivers are indispensable parts of the patient's recovery and reintegration, too.

While some of these professionals may only be part of the patient's team for a short time, the physiatrist and other rehabilitation professionals may be lifelong partners. Physical medicine and rehabilitation is uniquely positioned to help lead and coordinate this team, and it is their responsibility to best align the patient's medical recovery to the context of a broader life. Where possible, a physiatrist should begin evaluating and educating the patient prior to surgery, allowing the patient to set goals and expectations for recovery. In some cases, the rehabilitation team may consult with and advise the surgical team on ways to optimize postoperative function and prosthetic fit, considering the patient's personal, professional, and recreational activities, prior level of function, and comorbidities. After surgery, the physiatrist's long-term goals are to maximize performance and decrease pain. Portions of the ongoing care will include referring for therapy and other services, evaluating and prescribing prosthetic devices or other equipment, managing pain, and preventing and monitoring complications. Amputees will experience several stages of care, including surgery, postoperative care and acute rehabilitation, and advanced stages of rehabilitation. We discuss each stage of this process in the next sections of the chapter.

Surgical Considerations

Presurgical Evaluation

When evaluating the patient prior to surgery, the physiatrist should introduce the concept of the interdisciplinary team (see earlier). Education about amputation should be given, and expectations and goals should be established for the timeline of recovery. Almost all patients will not leave the hospital with a prosthesis, and most patients should plan for this process to take weeks to months (see later). Explain to the patient what to expect from surgery, acute rehabilitation, and introduce the concept of the rehabilitation team, with different team members available to help the patient succeed after limb loss. At this stage, collecting a relevant psychiatric history will help to anticipate the patient's likely needs upon discharge and determine candidacy for an eventual prosthesis. We recommend delaying in-depth discussion of prosthetic candidacy until after surgery for most patients, by which time acute medical circumstances have stabilized and patients have been evaluated by occupational and physical therapists.

Rehabilitation and Surgical Planning

Rehabilitation physicians should be aware of the surgical objectives and techniques frequently used in this population. The general goals of upper extremity amputation surgery³² are shown in [Box 10.1](#).

Whether to pursue limb salvage or amputation, and at what length to amputate, depends on many variables and remains a subject of debate. It is difficult to predict individual patient outcomes, and there is no firmly agreed upon formula for recommending a level or time of amputation for every patient. Most physicians would ascribe to preserving as much limb as possible.

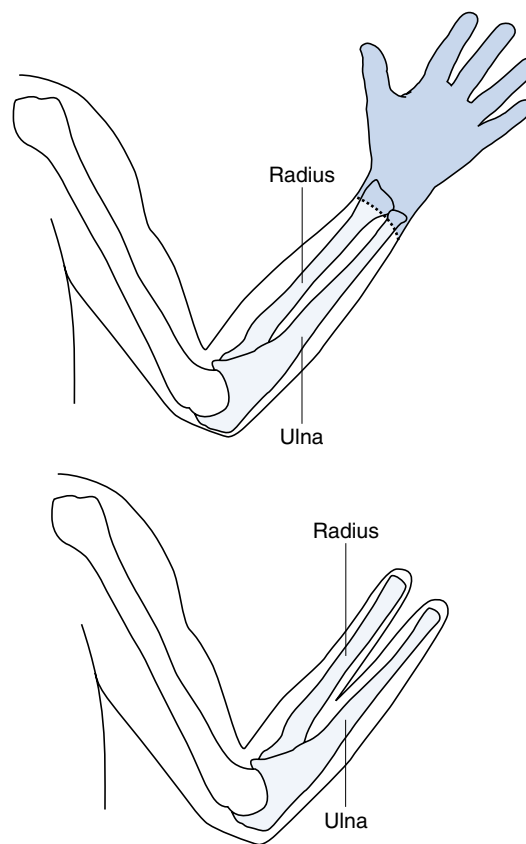
• BOX 10.1 Goals of Upper Extremity Amputation Surgery

1. Preserving functional length
2. Preserving useful sensation
3. Preventing symptomatic neuromas
4. Preventing adjacent joint contractures
5. Shortest and mildest morbidity possible
6. Early prosthetic fitting
7. Expedient return of the patient to vocational and avocational activities

A long transradial or long transhumeral amputation (preserving >60% length of the respective bones) generally provides better function and prosthetic use than a more proximal amputation through the same bone. Because the patient may already have traumatic or vascular compromise of tissue, the functionally ideal length is not always surgically feasible. In determining the level of the amputation, physicians consider the condition of nearby soft tissues, bone defects, proximal joint function, vascular status, and injuries to the contralateral side. If remaining soft tissue is insufficient to cover the residual limb or would be under excessive tension, a higher level of amputation may have improved outcomes. When there is injury to the bilateral limbs, every effort should be made to salvage the less injured side for a better functional outcome, even if it requires excessive shortening of the residual limb. Attempting limb salvage may also give the patient more time to make informed choices about further amputation procedures after the acute trauma and usually does not preclude the patient from undergoing a later amputation. For these reasons, limb salvage may be pursued even if complete success is unlikely.

A longer residual limb usually lends greater strength and range of motion, but it may not be ideal for every patient. For example, a wrist or elbow disarticulation preserves more length than a more proximal surgery, but a new prosthetic joint cannot be located at the same position as the original anatomic joint. This creates asymmetric limb length, poor prosthetic cosmesis, and, in some cases, reduced function and limited prosthetic componentry choices. Despite these drawbacks, an elbow disarticulation does usually provide better control of a prosthetic and better muscle attachment than a transhumeral amputation. While some patients may prefer this, others may prefer a more inconspicuous transhumeral prosthesis. Open and frank discussions are paramount in this decision. In dysvascular or infected patients, risks of poor healing or further tissue loss must also be considered against the potential benefits of attempting to preserve a joint. Patient preferences and lifestyle goals should always be considered, and it is often the role of the physiatrist and other rehabilitation team members to bring these factors into the surgical discussion. As always, the best decisions are made by a fully informed patient under the advisement of the interdisciplinary team.

There are surgical alternatives to amputation in select patients, such as the replantation of a severed limb. While it is a rare procedure, limb transplantation is also an area of active research. Upper limb amputees with blindness, such as those injured by a blast trauma, may have very limited ability to functionally use a terminal device because of the need for visual feedback. This rare situation could be an indication for a Krukenberg procedure, in which the radius and ulna are divided, and muscle groups rearranged to form a pincerlike residual limb that retains tactile



• **Fig. 10.3** The Krukenberg amputation transforms the residual ulna and radius into digits that have significant forceful prehension and retained ability to manipulate because of preserved sensation.

sensation and functional grasp (Fig. 10.3). This procedure has also historically been performed in resource-poor settings even for visually intact patients who will not have access to prosthetic technology and need a functional option. Krukenberg patients can also be fit with cosmetic prosthetics for social settings. Reconstructive surgeries may be performed after amputation to optimize patient function. These include index pollicization, metacarpal lengthening, phalangization, and even toe-to-thumb transfer. Although further details of these procedures are beyond the scope of this chapter, rehabilitation physicians should be aware of these options to aid in advising the patient and the surgical team.

Surgical Techniques: Soft Tissue Management

Three primary techniques utilized in amputation surgery for closure of the residual limb are myofascial closure, myoplasty, and myodesis. Myofascial closure involves approximating the outer fascial envelope over the muscles to close the limb. This approximates only loose tissues and does so under minimal tension to the underlying muscles. This is commonly thought to provide some benefit to healing in dysvascular patients, but this theory has limited empiric evidence. Myoplasty is the connection of opposing muscle groups to cover the transected bone end. This provides greater stability than myofascial closure but can cause pain with movement of these muscles as well as potential functional and prosthetic difficulties. Myodesis is the attachment of the residual limb's muscle groups to the end of the bone, either through drill

holes or through connection to the periosteum. This technique takes the most surgical time but prevents motion over the bone end and provides padding between the bone and a new prosthetic socket. In transradial amputations, for example, multiple muscle groups are available to perform myodesis to bone. This technique maximizes the distal soft tissue padding and control of the residual limb. While most amputations require at least partial transection of the muscle belly, in some cases a tendon may be available to attach to bone instead. This *tenodesis* provides the highest quality of muscle stabilization due to the good purchase of suture into bone with the tendon tissue.³²

Surgical management of skin tension is of paramount importance for healing and for functional use of a residual limb. The end of the residual limb must be adequately covered and padded with tissue. Efforts are made to avoid the formation of adherent, thin, and scarred skin covering, which is prone to injury and slow healing. In some traumatic and infected limbs, there may be multistage surgeries, where a first closure is allowed to heal partially and then later finalized. This may include skin traction, in which the wound is partially closed over a vacuum-assisted closure device (commonly called a wound vac) and allowed to begin healing. By maintaining tension over the end of the bone, muscle and cutaneous flaps are prevented from retracting as the limb heals. Infected wounds may require multiple debridement procedures to fully excise necrotic tissue and preserve as much healthy tissue as possible. Later, primary closure is performed with typical myodesis and tenodesis techniques. Skin grafting may be used to repair defects or provide viable tissue to complete a covering. Other concomitant injuries may complicate surgical planning and functionality, such as joint contracture or extensive burns.

Surgical Techniques: Nerve Management

Neuromas are a common complication following amputation. Neuromas are disorganized overgrowths of nerve tissue that form in the majority of amputees after a nerve transection. These neuromas may or may not be symptomatic, and who will develop symptomatic neuromas is currently impossible to predict. Symptomatic neuromas can result in both residual limb pain and phantom limb pain.

Nerve management in amputation surgery is a primary objective for long-term success. The traditional nerve management technique is a traction neurectomy, in which nerves in the amputated limb are placed under moderate tension and then transected. The nerves are then allowed to retract proximally into the wound and muscles of the residual limb. This technique accepts that a neuroma is almost certain to form but moves it away from high-pressure areas where it is more likely to become irritated and symptomatic. Other techniques place the cut ends of nerves into muscle tissue or bone near the end of the residual limb. In some patients, however, symptomatic neuromas still develop. The latter technique results in neuromas close to the distal end of the residual limb and can predispose to socket pressure and irritation in a prosthesis.

Recent advances in surgical technique are changing many practices and may improve outcomes. Two conceptually related procedures are targeted muscle reinnervation (TMR) and the regenerative peripheral nerve interface (RPNI). These procedures were originally developed to give patients more control over myoelectric prostheses but were also found to improve pain outcomes in amputees and reduce the incidence of neuroma formation.¹¹

Surgical details are beyond the scope of this chapter, but both techniques involve rewiring the limb's transected nerves into remaining neuromuscular targets, thereby reducing the chance of neuroma formation and pain. They can be performed at the time of amputation or later as a secondary surgery. Availability of these procedures is highly surgeon dependent, but based on patient outcomes, these techniques are likely to become more common. TMR is a procedure in which the transected ends of nerve from an amputation are anastomosed to motor entry points in remaining muscles. This differs from previous techniques by supplying the transected nerve with viable perineurium and an active muscle to innervate. This may reduce the development of symptomatic neuromas and pain.^{6,25}

TMR can be performed primarily at the time of surgery or secondarily at a later time as a treatment for symptomatic neuromas or phantom limb pain. RPNI is a newer technique in which transected nerves are implanted into small free muscle flaps, which are harvested from elsewhere. This provides new localized myoelectric signals for myoelectric prosthesis control. Based on available data, RPNI appears to have a similar effect to TMR on reducing neuroma incidence and phantom pain.²² The procedures were originally designed to improve myoelectric prosthesis control by providing additional myosignals. This can be especially useful for patients with proximal upper extremity amputations who need to control complex prosthetics with a limited number of remaining muscles.

Another recent innovation is the agonist-antagonist myoneural interface (AMI) procedure. Although the details are beyond the scope of this chapter, AMI can provide patients with improved prosthetic control and bidirectional signaling between their limb and a prosthetic device.²³ Human trials have shown potential for this to improve key skills, such as prosthetic control, sensory feedback, and proprioception.⁴¹ These and other promising procedures may enhance the abilities of patients with limb loss and prosthetics.

Surgical Techniques: Osseointegration

Osseointegration (OI) is a recent advancement in amputation care. It involves the implantation of a prosthetic interface directly into the bone of a residual limb, with the prosthetic attachment point penetrating through the skin. Once healed, this implant provides the direct connection of the patient's skeleton to a prosthetic limb. These patients do not use a socket and as such may experience improved sensory feedback known as osseoperception.¹⁶ OI is most useful in a patient who cannot tolerate socket wear or when a short residual limb makes it overly difficult to fit a prosthetic socket. Implantation of the device commonly requires two surgeries as well as extensive rehabilitation and time for healing before use. OI has been more extensively studied in the lower extremities, and guidelines for the upper extremity are less clear. This procedure is not currently US Food and Drug Administration (FDA) approved for the upper extremity. In the lower limbs, OI also comes with significant lifetime limitations on the use of the limb, such as the inability to jog, run, or carry heavy loads (cycling is still permitted). Risks of infection or implant-related fracture are notable concerns but are often preventable or treatable. Upper extremity patients may still seek out surgical specialists for this procedure abroad or through research programs within the United States. Anecdotal outcomes so far have been favorable, and it is likely that this will become increasingly common.⁸

Elective Amputation

Elective amputation may be pursued in rare cases where an intact limb is severely limiting function or quality of life. Limbs that are severely painful, severely contracted, paralyzed, or otherwise non-functional may prompt patients to seek out this option. Even if the patient may functionally benefit, it is important to ascertain if the patient's expectations of prosthetic restoration are realistic. This is especially true in the upper extremity, where replacing the vast capabilities of a hand is not yet possible even with the most advanced prosthetics. Patients frequently overestimate the capabilities of upper extremity prosthetic devices and underestimate their drawbacks. They may have an unrealistic idea of which prosthetics they will use and have access to. Surgery itself carries inherent risk, and finding a surgeon willing to sacrifice an otherwise viable limb may be challenging. If the patient is motivated by chronic pain, amputation is not a guarantee of pain control and may in some cases worsen the symptoms. Psychological evaluation is critical in cases where patients are considering such significant elective surgery. These cases are patient specific, and the risks and benefits need to be deeply understood by all team members.³⁸

Residual Limb Revision

Revision surgery can be performed in patients with prior amputation to address several issues. A surgical consultation may be indicated if a patient is experiencing pain or skin breakdown that has not responded to conservative therapies or prosthetic modifications. Additionally, the residual limb may be a suboptimal shape for prosthetic fitting, and revision might improve function. Bony prominences or heterotopic bone growth can be shaven or beveled to reduce their effect on the limb. Excess scarring or soft tissue can be revised to improve prosthetic fit. As described previously, neuroma pain can be addressed through a variety of techniques. Other specialized procedures may be able to improve function in select patients, such as the Marquardt humeral angulation osteotomy, which creates a bony angulation at the terminal end of the residual humerus to create what is best described as an artificial humeral condyle. This allows for better prosthetic suspension and more rotational control of the residual limb than in a typical transhumeral amputation.

Perioperative and Postoperative Management of Amputation

Once the surgical amputation is completed, the rehabilitation team stays in close contact with the patient and family if present. This initial phase of recovery can be quite variable, depending on medical stability and tissue health during surgery. These factors contribute to the final decisions for amputation length, soft tissue management, and skin and flap closure. Patients may also receive an initial guillotine amputation, which is a straight cut through the limb, to immediately address the necrotic and infected distal tissue. This is temporarily dressed, and after a period of recovery and stabilization, the patient returns to the operating room so that the amputation can be formally completed and closed. As discussed earlier, many patients undergoing amputation are doing so on an urgent or emergent basis. Due to the uncertain nature and comorbidities present at the time of surgery, it is not uncommon for patients to require a short- to medium-length intensive care unit admission for medical stabilization. A consulting rehabilitation

• BOX 10.2 Immediate Goals After Amputation

- Promote wound healing
- Control pain
- Control edema
- Prevent contracture
- Initiate remobilization and preprosthetic training
- Manage expectations through supportive counseling
- Guidance on rehabilitation pathway
- Continue education, including orientation to prosthetic components

team can be quite helpful in this setting, in assisting with splinting and bracing, therapy guidance, pain management, and early equipment recommendations. In general, the focus of the immediate postamputation period is to control pain and edema, promote wound healing, prevent contractures, initiate remobilization, assist with choosing a rehabilitation pathway, and continue the supportive counseling and education (Box 10.2). This plan must be individualized to meet the needs of each patient.

Postoperative pain control often requires an early, aggressive approach that considers the multiple potential pain generators in the postsurgical period. Pain control is discussed in more detail in the subsequent chapter on lower extremity amputation.

Edema control begins once the last suture or staple is placed by the surgeon. If there is no contraindication and the surgeon has the appropriate training, an immediate postoperative rigid dressing can be placed in the operating room. This is a special cast placed on the residual limb by the surgeon or another certified professional. Using the rigid dressing immediately for control of edema leads to earlier wound healing and improved pain control due to a reduction of pain mediators in the accumulated interstitial fluid. Depending on the amputation level and length of rigid dressings, they may also help prevent contractures of the nearby joint.⁵ Drawbacks of rigid dressings include a decreased ability to observe the wound, interference with drains and lines, and, if applied incorrectly, a risk of pressure injuries of the residual limb. These issues, combined with the experience required to place the dressings, means that many institutions do not regularly use them. This can be an excellent opportunity for the rehabilitation team to work with the surgeons to help improve surgical outcomes.⁴³

If rigid dressings are not an option, then soft dressings that assist with mild compression may be used. Garments or sleeves with a moderate-high compression pressure, such as prosthetic liners and shrinkers, should be avoided at this time. If a compressive dressing or wrap is used, it is important to be evenly distributed along the limb length to avoid any focal areas of pressure or strangulation points immediately distal to joints.

Passive ranging of the joints in the limb should be done daily during recovery to maintain mobility. If joints are noted to be losing range in the postoperative period, then increasing the frequency of ranging and/or utilizing appropriate bracing and splinting should be considered. If the patient has preexisting increased muscle tone or contractures, then more specialized care may be indicated to help prevent worsening of these issues.

Assistance with recommending a rehabilitation pathway should also be an ongoing conversation with the patient and primary team. A hospital-based team of therapists, including physical and occupational therapists, should see the patient as soon as possible after surgery for initial assessments and goal setting. Options such as admission to an acute inpatient rehabilitation

facility tend to be less indicated in upper extremity amputations, as opposed to lower limb amputations; however, the patient's needs, prior and current functional levels, and support network and resources should all be considered. Patients with an existing contralateral hemiplegia from pathologies such as stroke or spinal cord injury, or similar functional deficits of the remaining upper extremity, could easily warrant an acute rehabilitation admission. Patients who lose their dominant arm, especially if living alone prior to the amputation, may also require significant retraining to return home safely. In most cases, patients will be able to discharge directly home after medical stabilization and will follow up with the surgical and rehabilitation team on an outpatient basis. Early outpatient therapy will progress with an emphasis on recognizing the abnormal postures and positioning that occur with basic activities of daily living (ADLs).

ADLs are retrained with one hand, and patients use adaptive equipment when appropriate. The amputee progresses from independence with basic hygiene to advanced homemaking tasks. Hand dominance is retrained, when necessary, especially with handwriting and typing. Repetitive tasks can be used for strengthening. These tasks include fine motor exercises and gross motor exercises with equipment and mirrors. Isometric exercises are effective in creating muscle bulk for stabilization of the arm in the socket of the prosthesis. Because balance is often disrupted in a new amputee, the goals should include strengthening of the trunk, core, and lower limbs using isometric exercise and aerobic training. Depending on the level of loss, the upper limb amputee should begin to practice several motions that will be needed to control the prosthesis. Additionally, the stability of the prosthesis depends on both the bulk of the stabilizing musculature and the amputee's ability to voluntarily vary residual limb configuration.

Regardless of the rehabilitation pathway, early therapy should emphasize preserving range, strength, and maximizing use of the residual limb. Similar to the learned nonuse found in poststroke patients, upper extremity amputees may develop an early tendency to almost exclusively use their remaining sound arm. These habits can lead to less prosthetic success and a significant reduction in general functional potential.

Upper Limb Prostheses

An upper limb prosthesis is any device that replaces a portion of any upper limb that is missing; it can range from an artificial half finger worn solely to improve cosmesis to an electrically powered, computer controlled robotic arm meant to closely emulate an entire limb (Box 10.3).

Each prosthesis is unique and customized to the individual. Although two prostheses might be alike or very similar, there is no such thing as a normal or standard prosthesis. There are four broad categories of upper limb prosthetic systems: passive, body powered, myoelectric, and hybrid, which combines two or more of the others. The abilities of each system, and the requirements to use them, can vary significantly. The functional and vocational goals, the anticipated environmental exposures (such as sand, heat, or water), access to a prosthetist to build and maintain the device, and the financial resources of a patient all factor into prosthetic selection.

Passive or Cosmetic Prostheses

Some prosthetics are designed specifically to replicate the appearance of the lost limb and provide unobtrusive cosmesis without

• BOX 10.3 Prosthetic Terminology

- *Residual limb*: The remaining portion of the amputated limb
- *Excursion*: Amount of movement or range of motion that the residual limb can achieve
- *Component*: A part of the prosthesis
- *Terminal device*: The prosthetic equivalent of the human hand
- *Hinge*: A prosthetic component used to assist or replace an anatomic joint
- *Rigid hinge*: Stiff/solid arrow movement in only one plane, usually flexion and extension
- *Flexible hinge*: Allows movement in multiple planes (i.e., flexion, extension, pronation, and supination)
- *Socket*: This part of the prosthesis acts as the interface between the residual limb and the prosthetic device as a whole. The socket is designed to distribute forces throughout the residual limb.
- *Prosthetic sock*: A prosthetic component, usually cotton, that fits the residual limb like a sock and is worn between the socket and liner to account for volume changes (size changes) in the residual limb
- *Laminate*: A plastic composite usually made with carbon fiber and resin
- *Gel liner*: A prosthetic component made from a silicone gel or similar polymer that rolls onto the residual limb like a sleeve and creates a suction interface between the skin and the socket
- *Mechanical*: Designates a moving component
- *Passive*: Describes a prosthesis used only for cosmesis rather than function
- *Heavy duty*: Connotes a prosthetic designed to withstand strong, repetitive forces and rugged conditions (e.g., a heavy-duty prosthetic user could be one who works as an automotive mechanic or construction worker)

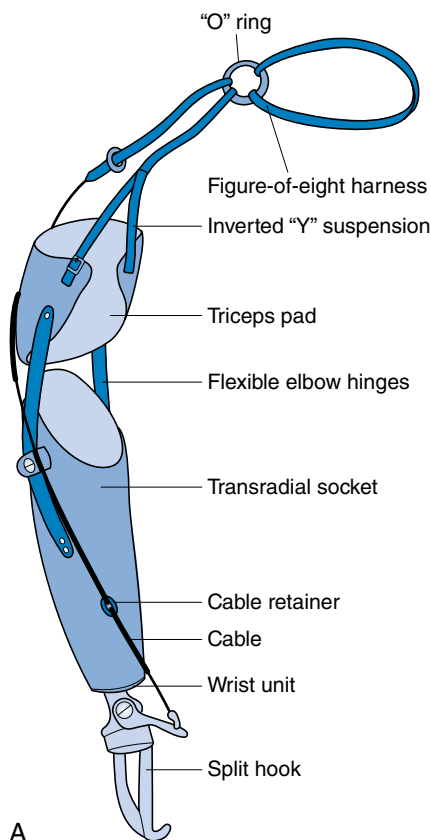
powered movement. These “passive” devices may still have joints or flexibility and are used functionally. They may act as a stabilizing weight or be used as a point of contact to move objects or pinch objects between the passive device and the sound limb. They are generally custom matched to the individual wearing them, including artificial nails, scars, and other skin changes to look more lifelike.

Body-Powered Prostheses

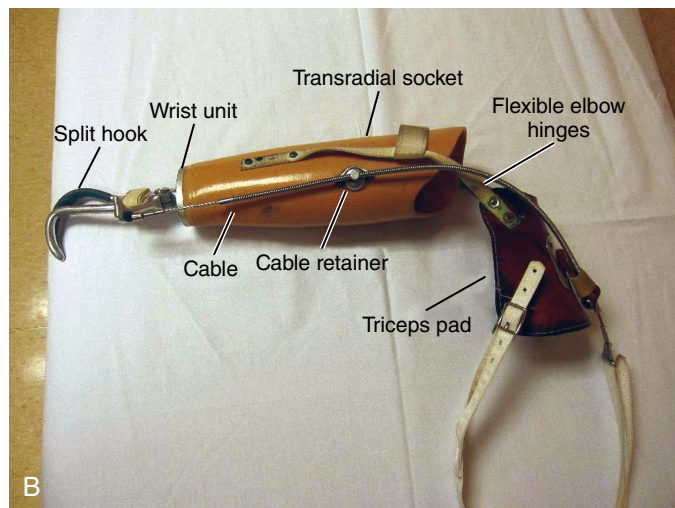
A body-powered system prosthesis uses the patient's own residual limb or body strength and joint movement to control the prosthesis (Fig. 10.4). Control includes powering the basic functions of terminal device opening and closing, elbow movement, and shoulder joint mobilization. The device may use the strength of the operating joint or body part to control the prosthetic joint, or it may use the body movement as a signal to externally power a joint. Most externally powered joints are battery operated to accommodate the freedom and mobility desired by patients.

Myoelectric Prostheses

Myoelectric systems use the electrical activity generated by the voluntary contraction of a skeletal muscle to trigger the externally powered movement of a prosthetic joint (Fig. 10.5). Myoelectric controls use sensitive sensors that are carefully located superficial to key muscles, commonly the wrist extensor and flexor compartments of the forearm, or the bicep and tricep muscles of the upper arm. Well-controlled and properly timed contractions of these muscles can be challenging and require significant amounts of therapy and practice to use successfully. Different types of myoelectric control systems exist. The two-site/two-function (dual-site) system has separate electrodes for paired prosthetic activity, such as flexion/extension or pronation/



A



B

• **Fig. 10.4** Diagram (A) and actual example (B) of a body-powered prosthetic control system.

supination. This control system is more physiologic and easier to control. When limited control sites (muscles) in a residual limb are available to control all the desired features of the prosthesis, a one-site/two-function (single-site) system can be used. Switch-controlled systems use small switches instead of sensors to operate the electric motors. These switches are typically enclosed inside the socket or incorporated into the suspension harness of the prosthesis, such as the “nudge,” which is operated by the chin depressing the switch on the anterior chest strap. A switch can also be activated by the movement of a remnant digit or part of a bony prominence against the switch or by a pull on a suspension harness.

Hybrid Prostheses

A hybrid system uses both the patient's own muscle strength and joint movement as well as an external supply for power. An example of a hybrid system is a body-powered elbow joint with an externally powered terminal device (Fig. 10.6).

Components: Socket and Suspension

There are multiple varieties of prosthetic sockets (Fig. 10.7). The most appropriate choice depends on the characteristics of the residual limb, the type and purpose of the planned prosthetic components, and the goals of the user. Self-suspended sockets benefit from strategically placed pressure points and bend to support the prosthesis wholly from the residual limb. Self-supported sockets tend to fit tightly and can require the use of donning aids, such as lubricants, to fully insert the residual limb. Harnesses can be used in place or in parallel with self-suspended styles of sockets to offload weight and gain support and stability from the remainder of the upper body. Two popular styles are a figure-of-eight (Fig. 10.8) and a figure-of-nine that loop around both shoulders or only the contralateral shoulder, respectively. Body-powered prostheses may have thicker and/or additional straps to serve as attachments for the cables that operate the prosthetic joints. Care must be taken with harness design because the pressure from the straps can damage skin and soft tissues and compress nerves and blood vessels.

The trim lines of the upper extremity socket are dependent on the length of the residual limb. The shorter the limb, the higher the trim line. For a longer limb, the trim line is lower and allows more pronation-supination, especially in transradial users. The patient's range of motion will be limited by a transradial prosthesis to approximately 70% of the motion possible without a prosthesis.

When the elbow joint is absent, the length of the transhumeral residual limb is a key factor in fitting and successful use of the prosthesis. Prosthetic control varies directly with the length of the humerus. Amputation through the distal third of the humerus provides functional control very similar to an elbow disarticulation. Transection of the humerus at least 4 inches above the olecranon allows enough clearance to use all elbow options, including externally powered.

Forequarter-level amputations (loss of scapula, clavicle, and chest wall muscles) present even more of a challenge to fit with a functional prosthesis. Most control options have been removed with the residual limb. The “nudge” control, which is a force-sensitive resistor operated by the chin, can operate the elbow and hand. If no prosthetic is used, a cosmetically sculpted insert should be offered for the shoulder symmetry needed for clothing fit.

Components: Shoulder

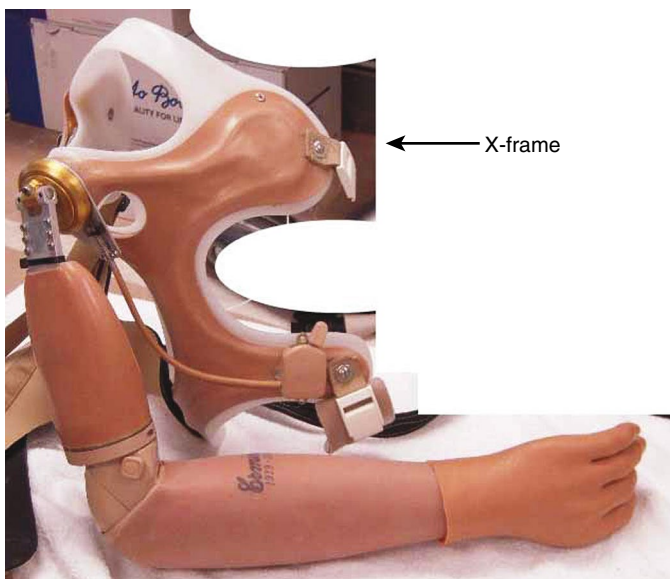
Shoulder joints are rarely used in prostheses due to the additional degrees of freedom and the greater control input required for meaningful use. Body-powered shoulder joints are more common, although cutting-edge prostheses may utilize an externally powered shoulder joint, often operated by foot pedals or other switches.

Components: Elbow

Elbow joints may be body or externally powered, with the choice often driven by the preference for terminal device control and the



• **Fig. 10.5** Myoelectric prosthesis (A) with myoelectric control system (B) for a transradial amputee with socket electrodes visible.

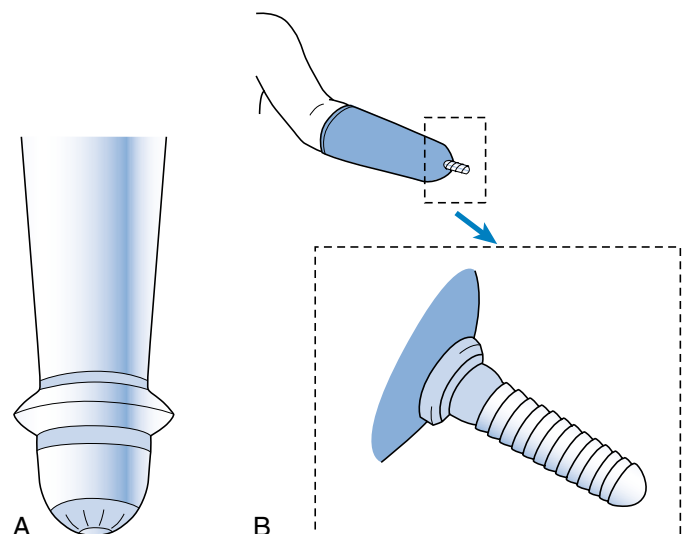


• **Fig. 10.6** Shoulder disarticulation and forequarter prosthesis, body-powered elbow, and externally powered wrist and hand.

presence or absence of a residual limb shoulder joint. In transhumeral users with a fully body-powered prosthesis, it can be challenging to operate both the elbow and terminal device. Often the wrist joint is manually operated to reduce the amount of required input. Still, the prosthesis may require specific sequencing of the residual limb and torso to operate a specific joint in the desired way.

Components: Wrist

The wrist unit provides orientation of the terminal device in space. It can be positioned manually, by cable operation, or with

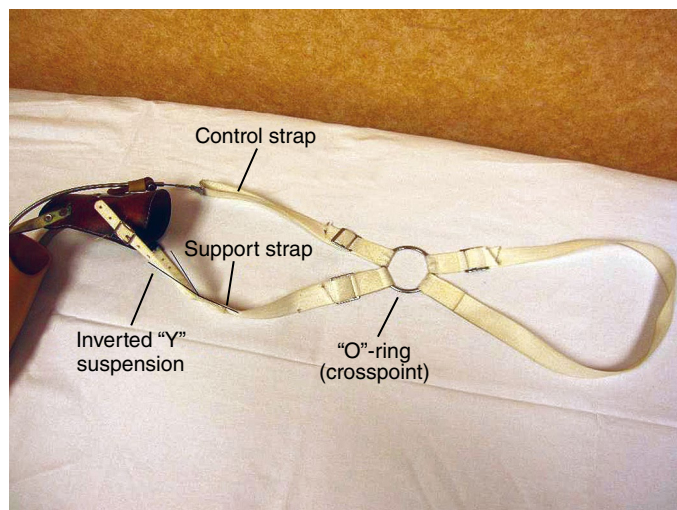


• **Fig. 10.7** Examples of prosthetic suspension systems. (A) Suction style with a seal-in ring. (B) Distal locking pin style, which inserts into a corresponding lock with quick-release button.

external power (myoelectric or by switch). Once positioned, the wrist unit is usually held in place by a friction or mechanical lock.

Components: Terminal Devices

There are many terminal devices available for upper limb amputees, although the functionality of these terminal devices tends to be limited. Terminal devices are generally one of two categories: passive and active. Passive terminal devices fall into two classes: those designed primarily for function and those that provide cosmesis. Examples of the functional passive terminal devices include the child mitt frequently used as an infant's first prosthesis to



• **Fig. 10.8** Components of the figure-of-eight harness: axilla loop, anterior support strap, control attachment strap, and crosspoint.

facilitate crawling and the ball-handling terminal devices used by older children and adults for ball sports.

Active terminal devices are also one of two main categories: hooks (including prehensors, which are devices that have a thumblike component and a finger component producing a claw or bird's-beak type of function) and artificial hands (Fig. 10.9). Both device groups can be operated with a cable or by external power. No single device can reproduce the complex functional capability of the human hand.

Cable-operated terminal devices (hooks or hands) can be of a voluntary opening design (most common) or a voluntary closing design. With a voluntary opening mechanism, the terminal device is closed at rest. The patient uses the control-cable motion to open the terminal device against the resistive force of rubber bands (hook) or internal springs or cables (hand). Relaxation of the proximal muscles allows the terminal device to close around the desired object. The number of rubber bands determines the

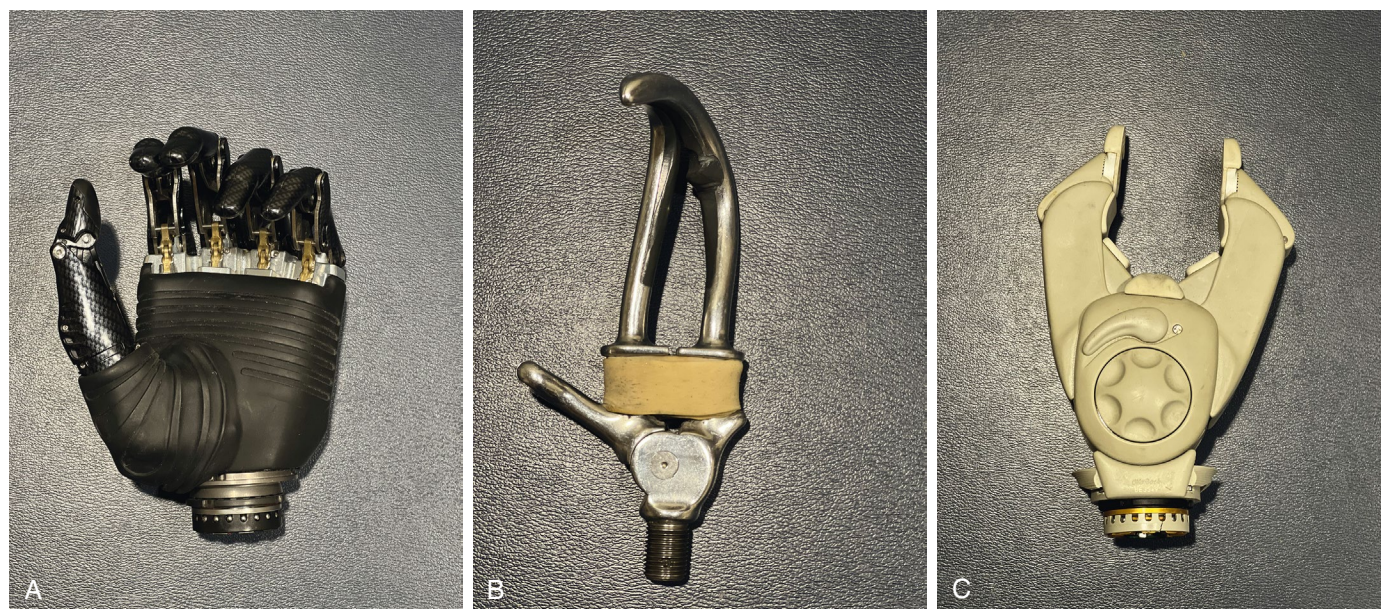
amount of prehensile force that is generated. One rubber band requires 5 pounds of force to provide approximately 1 pound of pinch force (pinch force in a nonamputee is typically 15–20 pounds).

Hooks have many advantages: They are simple in design, lightweight, and durable; enable efficient grasp; have low maintenance; and permit visual feedback that is unavailable with a mechanical hand.

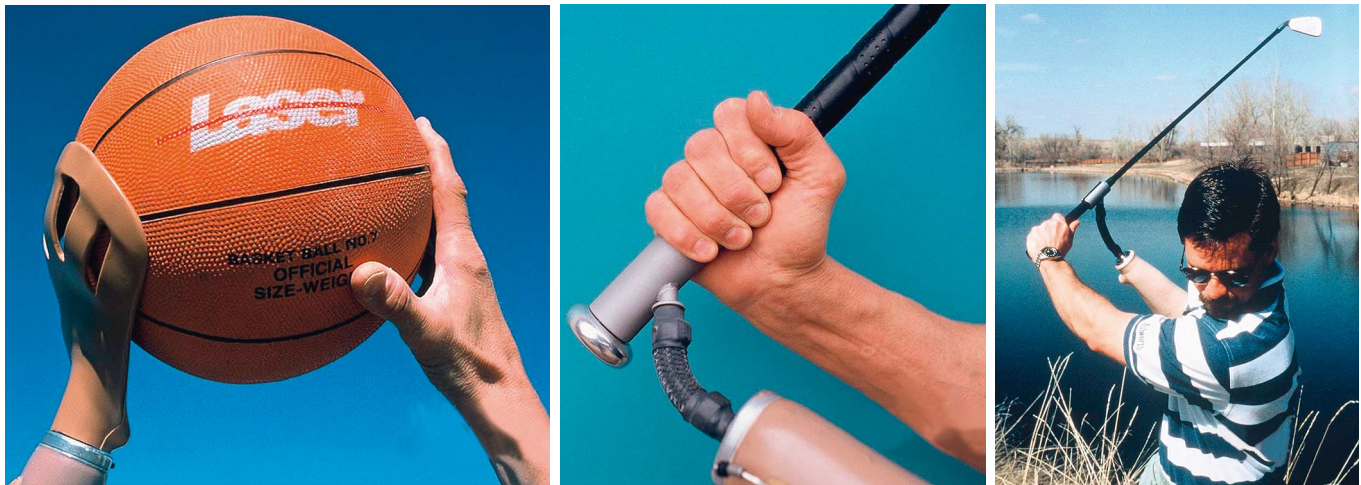
A prosthetic hand usually is bulkier and heavier than a hook. It can be powered by a cable or use external power. Some of the current hands on the market have unique grasping characteristics. These include a feature used to eliminate slipping of an item being grasped. Most hands are capable of several grips; however, switching between grip styles can be challenging, and most amputees tend to favor one or two programmed grips. Examples of the five major grip types can be found in Box 10.4. Not all electronic hands look like natural hands. Hands made for heavy-duty industrial purposes have other hooklike shapes. Task-specific terminal devices or systems also exist, such as a “basketball hand” or a bat/club attachment (Fig. 10.10) that may be used in conjunction with the abovementioned systems or deviate enough to earn its own category.

• BOX 10.4 Five Major Types of Grip

- *Precision grip* (i.e., pincher grip): The pad of the thumb and index finger are in opposition to pick up or pinch a small object (e.g., a small bead, pencil, grain of rice).
- *Tripod grip* (i.e., palmar grip, three-jaw chuck pinch): The pad of the thumb is against the pads of the index and middle finger.
- *Lateral grip* (i.e., key pinch): Tips of the fingers and thumb are flexed (e.g., when screwing in a light bulb or turning a doorknob).
- *Hook power grip*: The distal interphalangeal joint and proximal interphalangeal joint are flexed with the thumb extended (as when carrying a briefcase by the handle).
- *Spherical grip*: Tips of the fingers and thumb are flexed (e.g., when screwing in a light bulb or opening a doorknob).



• **Fig. 10.9** (A) Myoelectric artificial hand. (B) Body-powered hook. (C) Myoelectric Greifer (OttoBock).



• Fig. 10.10 Activity-specific terminal devices.

The Early Prosthetic Phase

Prosthetic Evaluation and Goals

When a patient with limb loss is being considered for a prosthesis, there are three initial areas of focus for the physiatrist.

First, has the residual limb fully healed from the surgery? Are there other new wounds or lesions since amputation? As in the lower extremity, adding a prosthesis adds pressure, heat, and weight to the underlying skin, thereby increasing the risk for damage and worsening existing skin issues. Has the patient already been prescribed a residual limb shrinker to help decrease tissue edema and shape the limb in preparation for a prosthesis? If not, the patient should be able to tolerate the shrinker for 23 of 24 hours a day, removing for personal hygiene and cleansing of the shrinker.

Second, is pain present, either residual limb or phantom limb pain, that will interfere with potential prosthetic use? Pain can be treated concurrently during the prosthetic fitting and fabrication stages, but if wearing the final prosthesis causes pain, then there is a high potential for device abandonment.

Third, and perhaps most importantly, what are the goals of the potential prosthetic user? Understanding a patient's current capabilities, goals, and intended use of the artificial limb is paramount to providing a prosthesis that is desirable and useful. No artificial limb will be a true replacement for what was lost. Prosthetic limbs are tools used to help accomplish specific goals and tasks. Especially with upper extremity loss, having a frank conversation with the patient helps to set expectations and ultimately influence the acceptance and use of any provided limb.

Once a patient's residual limb is ready for a prosthesis, pain is moderately to well controlled, and goals and expectations have been set in clinic, then a prosthetic prescription may be written. Like a medication prescription, there are certain required elements for the order. A specific diagnosis must be made (e.g., "Acquired absence of left upper limb above elbow Z89.222"), and the requested categories of components should be listed to include suspension type, socket type if needed, body powered or externally powered, and with or without myoelectric control if appropriate. Unlike lower extremities, upper limb prosthetics are not indicated based on a potential ambulation assessment or K level. The specific components and level of technology are normally determined based on the goals of the user and, ultimately, the insurance coverage and financial resources of the user. The

cost of an upper extremity prosthesis can start at \$3000 for a passive device, \$10,000 for a body-powered hook, and \$20,000 to \$40,000 or more for an externally powered myoelectric arm. Insurance, if present, may cover all the cost, part of the cost, or in some cases decline any reimbursement at all. While a patient's health, capabilities, and goals should drive the primary decision making, it is important to understand and offer guidance on the financial realities of prosthetic costs. For veterans in the United States, note that the Veterans Affairs (VA) Hospital System operates a robust nationwide amputee rehabilitation network, and any veterans with limb loss should be encouraged to reach out to their local VA medical center to explore potential resources.

In most situations, it is beyond the physician's knowledge and experience to fully prescribe the exact componentry of the prosthesis. Maintaining a clear line of communication with the patient's prosthetist, or even better having them present in clinic, provides optimal quality and efficient care to the patient. Once prescribed, the patient will work with the prosthetist to develop a plan for the componentry, suspension system, and prosthetic socket. Often a clear plastic "check" socket will be produced to optimize the fit and function of the prosthesis. This allows monitoring of the skin and soft tissues of the residual limb during wear. Once the fit is finalized, a definitive socket, typically made of a plastic composite with carbon fiber and resin, will be produced and mounted to the prosthesis. After initial provision of the prosthesis, it is important for patients to follow up with their prosthetist regularly to ensure optimal fit and function.

Ideally, a patient should follow up with the physician within 1 to 2 months after receiving the definitive prosthesis so that a "checkout" visit may occur. This allows the patient to report any issues, such as pain or skin breakdown, and confirm that the device is meeting current needs. It also allows unforeseen problems with the prosthesis to be examined, and new prosthetic orders written if needed.

Early Prosthetic Training

Training for the upper limb amputee should ideally begin before the surgery and continue until advanced training is completed. The training is divided into three phases: preprosthetic training, early prosthetic training, and advanced prosthetic training. Each phase is focused on the end goal of maximal functional adaptation and proficiency with the prosthesis. The expectations of the

amputee might not match reality once the prosthesis is delivered and initially worn. It is important to reinforce to the unilateral amputee that the prosthesis will play a nondominant but important functional role. It is a “helper.” The sound hand will always be dominant for all activities performed. Progressive attention to psychological adjustment and changing body image needs to stay at the forefront. Making use of the trained peer visitor is a powerful tool.

The prosthetic training phase begins with the delivery of the prosthesis. A key initial focus is on donning and doffing and wearing the prosthesis for short periods with gradually increasing duration of prosthetic wear. Integrating the prosthesis into daily activity is the chief goal. Numerous issues arise during this period of rapid change, including maladaptive habits that occur quickly, such as inefficient body movements for control. After the introduction to the prosthesis is completed, training progresses toward mastering basic ADLs (Box 10.5).

After ADLs are completed, the amputee is then moved to higher level homemaking skills and community reentry activities, such as driving, work, and recreation. Protocols for controls training for the body-powered prosthesis and the myoelectric prosthesis have been reported by Ganz et al.¹⁵ Additionally, Northwestern University has designed a group of activities and a rating guide¹ as a helpful tool for the therapist to use to guide and assess progress for the unilateral upper limb amputee (Table 10.1).

Proprioceptive neuromuscular facilitation enables the therapist to key into specific areas of muscle weakness. Isotonic exercises are effective in maintaining muscle bulk for stabilization of the arm in the socket of the prosthesis. The stability of the prosthesis depends on both the bulk of the stabilizing musculature and the amputee’s ability to voluntarily vary residual limb configuration. For the transhumeral level, this would be the external rotators and the biceps that stabilize the socket; for the transradial level, it would be the supinators and pronators that stabilize the socket. There is a natural linear flow of these three rehabilitation phases. As the amputee attempts to integrate back into life, prosthetic training advances to mastering more specific and unique tasks (such as those found in Table 10.1). The prosthetic training phase ends with the proficient use of the prosthesis. Ultimately, successful implementation of advanced prostheses will be largely contingent on the availability of highly specialized and trained clinical personnel to fit and train amputees and resources to pay for these services.³⁷

During prosthetic prescription and therapy, clinicians should consider using objective functional outcome measures to monitor

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BOX 10.5

Introduction to a Prosthesis

- Operational knowledge:* Basic vocabulary and knowledge of components, including the ability to communicate malfunctions
- Maintenance and care:* Include socket inspection, maintenance, cleaning, and adjustments
- Residual limb care:* Includes a wearing schedule to progress to full tolerance (~8 h/day) and learning routine daily inspection of the residual limb for irritation
- Controls training:* Includes learning to control the individual components for operation. The five motion elements that are primarily used in hand manipulation are reach, grasp, move, position, and release. The therapist uses many objects, such as wooded blocks, cotton swabs, and a sponge, in training the individual.

TABLE 10.1
Example of a Rating Guide

Criteria	Examples	Grade ^a
Personal needs	Don/doff pullover shirt Manage zippers and snaps	
Eating procedures	Cut meat Open milk carton	
Desk procedures	Use phone and take notes Sharpen pencil	
General procedures	Operate doorknob Set time on watch	
Housekeeping procedure	Hand wash dishes Dry dishes with a towel	
Use of tools	Hammer Tape measure	
Car procedures	Open and close doors, trunk, and hood Perform steps required to operate vehicles	

^aRating guide grading: 0, not possible; 1, mostly clumsy but accomplished; 2, minimal clumsiness; 3, smooth. (Grading system developed for individualized goals.)

a patient’s progress. Studied tools, such as the Patient Reported Outcome Measurement Information System (PROMIS) Upper Extremity Physical Function item bank, are relatively brief and were developed specifically to evaluate this population by the prosthetics company Hanger Clinic.^{13,20} The PROMIS tool is also available in a pediatric version.

Special Prosthetic Considerations

Bilateral Limb Loss

Bilateral amputees must use prostheses for all upper extremity activities. Artificial limbs can be much more challenging to don and doff, and care must be used in their design to help enable independence. Especially in heavy users, body-powered or hybrid systems with hooks tend to be favored by bilateral patients due to their durability, reliability, greater level of sensory feedback, and greater ease of donning/doffing. Additionally, prosthetic controls training is a more complex and coordinated motor process. The rehabilitation team needs to maintain and progress the strength and coordination gained in the pre-prosthetic training.

Burns and Skin Grafts

Amputations associated with trauma may also have damage to nearby soft tissues and skin or have skin grafts present after such damage. Skin grafts may take months to a year to fully mature, and it is important to communicate with the surgery team for discussion on prosthetic options and timeline.

Vision and/or Hearing Loss

Virtually all available prosthetic limbs lack sensory feedback and rely almost entirely on the user’s sight and hearing. When deficits

are present, especially in vision, prosthetic use can prove challenging or even impossible. In patients with severe visual impairment, surgical procedures are sometimes pursued to improve the use and dexterity of the remaining limb, such as the previously mentioned Krukenberg procedure.

Traumatic Brain Injury and/or Cognitive Deficits

In patients with the inability to understand or follow safety instructions, a prosthetic device may provide more risk than benefit. If a prosthetic is to be prescribed, its reliability and ease of use should be prioritized. Discussion and shared decision making with the patient and the family is critical to determining goals and prosthetic choices.

The Advanced Prosthetic Phase

Once patients' residual limbs have stabilized in size (which can take ≥ 1 year), swelling has reduced, wounds have fully healed, and they have begun to successfully use a prosthetic device, then they transition into the advanced phase of rehabilitation. The goals of advanced prosthetic rehabilitation focus on maintaining and optimizing function, minimizing pain, preventing common complications, and adapting to the patient's medical and lifestyle changes.

As a general guideline, patients with upper extremity amputation who have reached the advanced prosthetic phase should be followed up at least yearly. At these visits, the clinician should assess the basic domains found in [Box 10.6](#). Some of these needs that are specific to the upper extremity will be discussed in this chapter, and others will be covered in detail in the chapter on lower extremity limb loss.

While the need for regular lifelong follow-up by the rehabilitation team cannot be overemphasized, it is frequently underutilized. Without follow-up, many barriers to successful prosthetic use cannot be addressed, and functional use can be greatly diminished or absent. Issues such as pain, depression, skin irritation, limb size change, and changes in activity levels or use are all more easily addressed early. In some cases, serious complications, such as limb infection, falls, or prosthetic failure, can be averted through preventative care at these visits. Rehabilitation physicians should work proactively to prevent complications when possible. They play a key role in the health and safety of an amputee. In the case of an otherwise healthy, traumatic amputee, the rehabilitation physician may be the only regular point of contact with the health care system.

Patients should first be assessed for any significant medical or life changes since the last evaluation. Physicians should inquire

how well they are functioning in their work and personal lives and whether they have concerns or functional impairments that they would like to discuss.

Prosthetic fit is an important aspect of amputee care that frequently changes over a patient's lifetime. Volume change in the residual limb is common, especially within the first year after amputation. Upper extremities are less susceptible to volume changes than the lower extremities due to gravity, but nonetheless may require adjustments of the prosthetic socket or new choices of suspension. If a patient is experiencing significant changes throughout a given day, new suspension strategies may be indicated, along with wearing compression garments. Significant weight changes can severely impair prosthetic fit, especially in the upper extremity where sockets are fit very intimately to the patient's anatomy. It is critical to ascertain the medical reason for the patient's apparent weight gain because serious conditions such as congestive heart failure, renal and hepatic disease, venous occlusion, metabolic syndromes, or changes in medications such as diuretics should not be missed.

Pain Assessment and Management

Pain in the amputee population can be categorized as *residual limb pain*, originating from the intact portion of the limb, or *phantom limb pain*, originating from the amputated limb that is no longer present. Residual limb pain has a wide differential diagnosis and encompasses both amputation-related and general musculoskeletal and neurologic conditions. As discussed previously, neuromas formed after the amputation are common sources of both residual limb and phantom limb pain. Their clinical management is discussed in detail in the chapter on lower limb amputation. Surgical procedures mentioned previously in this chapter, such as TMR and RPNI, are emerging and promising management options. Musculoskeletal pathologies, such as arthritis, tendinopathy, bursitis, or ligament tears, may present as residual limb pain. It is important to recall that patients with amputations may also develop complex regional pain syndrome, and this should be included on any pain differential, particularly when pain appears neuropathic in nature with vasomotor or skin findings or when these symptoms are temporally delayed from the amputation event.¹²

Phantom limb pain is a unique problem to the amputee population. Clinically, it is best understood as a maladaptive neuroplastic response to the loss of the limb. In these cases, the brain interprets the limb's afferent nerve signals, or lack thereof, as pain. It has both central and peripheral mechanisms, which despite extensive research remain poorly understood.⁴² It is typically described as a neuropathic sensation, with burning, stabbing sensations that wax and wane, and is often worse at nighttime. Patients may have difficulty in clearly describing their pain. The prevalence of phantom limb pain in the general amputee population has been estimated at 64% in a recent meta-analysis, although estimates vary widely depending on methodology and population studied.²⁸ In most of these patients, symptoms are intermittent and may arise occasionally for seconds to minutes before resolving, and it may not significantly impair their functionality or quality of life. In some patients, however, phantom limb pain can become a chronic, severe, and debilitating disease that is notoriously difficult to treat. Treatment strategies for phantom limb pain of the upper limb are like those discussed for the lower limb in the following chapter. These include medical, therapeutic, and surgical management. Unfortunately, scientific evidence for best practice

• BOX 10.6 Domains to Be Addressed at Follow-Up Visits

1. Major medical or life changes
2. Function and prosthetic fit in work and recreation
3. Musculoskeletal and neuropathic pain
4. Skin condition
5. Health and condition of the sound limb
6. Psychosocial condition, need for social services
7. Reassessment of rehabilitation goals
8. Assistive device and equipment needs
9. Prosthetic resupply, replacement, or change

in this area is weak, and patients with severe symptoms frequently trial many types of treatment.¹⁴ It is important to differentiate phantom limb pain from the more common and less problematic phantom limb sensation, which is the nonpainful sensation of the limb still being present. The prevalence of phantom limb sensation is nearly universal after limb loss and often remains with the patient for life.

Musculoskeletal Comorbidities

Patients with upper extremity amputations are at higher risk than the average population for other common musculoskeletal problems, such as neck pain or rotator cuff pathology.⁴⁴ The management of most of these conditions is not unique to amputees and is discussed in other chapters. Patients should be screened for pain and dysfunction of all proximal joints and muscles to their amputation. Some patients may develop an amputation-specific bursitis, tendinopathy, or enthesopathy in their residual limb. Surgical clips and other retained hardware contribute to risk of pain near the operative site. The sound limb should also be screened for pathology, with particular attention for overuse injuries such as carpal tunnel or rotator cuff pathology. The optimal health of the sound limb is of paramount importance and should not be neglected during rehabilitation or follow-up appointments.

Heterotopic Ossification

Heterotopic ossification (HO) is the pathologic formation of bone outside of existing skeletal tissue. It can present with significant pain and forms within weeks or months after amputation. In upper extremity amputees, symptomatic HO can cause pain, skin breakdown, and ultimately may impair the use of a prosthesis. It is often palpable in the residual limb or detectable by x-ray or ultrasound. Triple-phase bone scan is not usually required unless x-ray imaging appears negative with high clinical suspicion. The mainstays of conservative treatment are activity modification, prosthesis adjustment to accommodate limb changes, and physical therapy to maintain range of motion.³¹ There is no available strong evidence for any medical intervention to slow the progression of HO in the amputee population. Nonsteroidal antiinflammatory drugs, bisphosphonates, and local radiation have been trialed as prophylactic and therapeutic options in other patient populations but have not been proven in the amputee population at this time.³³ Other medications that have been trialed but have limited evidence include colchicine, corticosteroids, and calcitonin.^{29,32} HO can be surgically excised. Classically, this is performed once it has fully matured, which takes roughly 6 to 18 months.¹⁹ However, it is now becoming more common to proceed with earlier excision as long as the patient has failed conservative measures, and the HO is corticated and well defined on imaging.³⁴ In extensive or nonresectable cases, amputation revision may be considered instead.

Dermatologic Management

Skin condition must also be assessed at each visit. Residual limb skin care and the management of common amputee skin issues are discussed in detail in the chapter on lower extremity limb loss. Skin irritation leading to blistering, skin breakdown, and topical infection are common, especially with poor prosthetic fit. This is most commonly due to a worn prosthetic suspension, such as a damaged gel liner, excessive use, or abrupt changes in wear time.

Other common etiologies include skin irritation due to skin creams, antiperspirants, and other topical lotions or medications. Patients should be encouraged to clean and moisturize their residual limb daily, but lotions should never be applied just prior to donning a liner or prosthesis. For soft tissue infections, topical and antibiotic treatment may be indicated, and prosthetic wear should be limited to encourage healing. Common skin conditions seen in amputees are discussed in greater detail in the chapter on lower extremity amputees, but include acroangiokeratosis, verrucous hyperplasia, and epidermoid cysts. Acroangiokeratosis is a condition that produces painful or pruritic purplish papules and plaques due to venous insufficiency and stimuli from socket wear. Verrucous hyperplasia is a condition resulting in wartlike overgrowth on the residual limb, usually caused by venous pooling in the distal end due to poor prosthetic fit.³⁵ Epidermoid cysts of the residual limb may also develop and can be treated with excision if symptomatic.²⁶ Skin infection of the residual limb is not uncommon. It is frequently caused by contact dermatitis blistering, pressure ulceration, or bacterial or fungal cellulitis. Prosthetic rest and hygiene can minimize this risk. Deeper infection, including septic bursitis, is also possible and typically presents with a swollen, red, and painful limb. Severe cases may require hospital admission to avoid limb compromise and sepsis.

Hyperhidrosis

Excessive sweating (hyperhidrosis) of the residual limb can impair prosthetic use and be distressing to the patient, particularly in hot climates or with high-activity demands. Prosthetic sockets can sometimes accommodate drainage or breathability, or additional socket layers may be used to absorb perspiration. Topical antiperspirant (preferably without deodorant to avoid skin irritation) is often an effective solution and is recommended to be applied at bedtime rather than in the morning to avoid skin irritation after donning a prosthetic. In refractive cases, botulinum toxin may be injected into the skin at areas of excessive sweating and, if effective, may last up to several months or more before needing to be repeated.

Rehabilitation Goals and Prosthetic Prescription

At each clinic visit, the patient and care team should reassess their progress and goals and determine if any change in course is needed. Are patients and the team satisfied with their progress? Have patients changed jobs or started a new hobby? Have they achieved their goals and potential? Are there barriers preventing this? Is there new technology or treatment that might benefit them? Some of these might be best addressed with occupational or vocational therapy or assistive/adaptive devices. Prosthetic equipment such as socks, liners, or components may be worn and need replacement. The frequency of replacement can change based on use and environment, but new liners and socks every 6 months is reasonable; it may be more frequent if the items are found to be worn. Changes in the patient's body or activities may necessitate adjustment of an existing prosthetic or the prescription of a new one to meet their needs. Note that activity-specific prosthetics may not be covered by most insurance plans depending on the indication. Many activity-specific subtypes of prosthetics, such as recreational limbs, water-resistant prostheses ("water legs"), and those made for cosmetic purposes are unfortunately not covered by most insurance due to being deemed not medically necessary. There is no firm rule on replacement of an upper

extremity prosthetic, and their lifespan varies greatly depending on use; however, general insurance guidelines allow for replacement of an entire upper extremity prosthesis once wear and tear has become significant enough that it impairs function and cannot be reasonably repaired or modified. Physicians may also determine that a patient's medical status necessitates a change in the prosthetic. Intolerable skin irritation, limb pain, neuromusculoskeletal comorbidities, vision changes, or decreased cognitive ability are all examples of when it may be medically necessary to change to a different prosthesis or discontinue its use.

Upper Limb Loss Adaptation and Reintegration

While a well-fitting prosthesis can provide significant utility, it is not a replacement for the lost limb. Even if ADLs are easily accomplished by the patient, it is likely that there are multiple aspects of their home, vocation, or hobbies that will require adaptation and modification. A consultation to a physical therapist or occupational therapist for a home safety evaluation, use of adaptive equipment, and home modifications must be considered. As in lower extremity amputation, the self-perception of a patient with limb loss can be severely altered or impaired, and early and regular visits with mental health providers can assist with adjustment and body image concerns. These psychological changes can also significantly affect relationships with spouses, families, and friends, and it may prove useful for those with partners to participate in couples counseling. In almost all cases of upper extremity amputation, the ability of the patient to work will be impaired. Consideration should be made for consultations for vocational rehabilitation, requests for accommodation at work, and ensuring a relationship with a social worker or case manager to assist with disability applications as appropriate.

Psychological Reintegration, Community Support, and Disability

Limb loss is a profound life change, and regular attention should be paid to patients' mood and feelings about their limb loss. Depression risk is elevated during the first 2 years after amputation in multiple studies,^{21,24} but little data exist for long-term follow-up. Psychological counseling or referral to psychiatry can also be helpful and should be considered part of regular follow-up.

The limb loss community is particularly active worldwide through support groups, advocacy, and media. Connecting patients to these resources can play an important supplementary role in their recovery. Local amputation support groups meet regularly in many midsize and large cities to connect patients with peers. The ACA is a very active national group that provides peer support, education, and advocacy for those with limb loss. The organization publishes educational items such as books and pamphlets (often in multiple languages and sizes of text) and holds large annual conventions for those with limb loss, their family and friends, and health care professionals. Other resources can be found through organizations such as the Veterans Affairs Health System or through sponsored events organized by nonprofits or prosthetic companies. There are also amputee-centric publications, such as *Amplitude Magazine*, that provide an educational and media outlet focused on this community. The educational materials available to patients through these groups span a wide variety of topics, from insurance navigation, legal advocacy, research developments, and resources to less discussed issues,

TABLE 10.2 Example of Disability Rating

Disability	Rating (%)
Loss of one upper limb	50
Loss of one hand	45
Thumb amputation	23 (50% of one hand)

including sex and intimacy. Interacting with these groups can provide an important sense of belonging and support during a period of intense psychological adjustment.

Many aspects of upper limb prosthetic rehabilitation cannot be addressed until the patient has had reasonable time to become acclimated to the life and functional changes being experienced. These include housing and employment concerns. At each follow-up, patients should be evaluated for potential social services referrals. The physiatrist may also be asked by insurance to state the patient's level of disability (Table 10.2).

The team should remain available to address questions and nurture newly mastered skills. The fit, comfort, and function of the prosthesis must be maintained and optimized over time as amputees alter and refine their initial goals and aspirations. The successful long-term use of an upper limb prosthesis depends primarily on its comfort and its perceived value to the amputee. Innovative design and careful custom adaptation of socket and harness, careful attention to follow-up adjustments, and prescription revisions based on the amputee's changing needs are the essential factors for successful prosthetic rehabilitation.

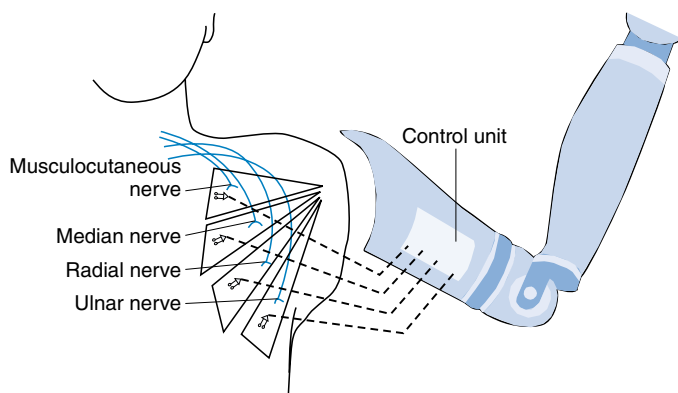
Emerging Technology

The sheer amount of new and developing technology could easily fill a textbook of its own. What follows is a short overview of trends and new areas of innovation.

1. Powered prosthetics. There have been motorized versions of all upper extremity prosthetic joints for many years. Most externally powered hands, however, tend to have motors in the thumb and first three digits only. Newer models can have motors in all digits, allowing individual movement of the fingers as well as flexion and opposition of the thumb. Even more advanced devices allow finger abduction and adduction.
2. Three-dimensional (3D) printing and digital design. Otherwise known as additive manufacturing, 3D printing allows an object to be built layer by layer from a material, typically a plastic polymer, from a digital design. In the future, 3D printing may allow patients who do not have access to a prosthetist to access custom-fit artificial limbs.
3. Pattern recognition. Traditional myoelectric limbs use one or two small sensors directly over muscles to capture muscle movement. Newer input devices can have eight or more sites spread circumferentially around a limb and use computers to recognize patterns as patients emulate certain movements. These patterns can be used as triggers for motorized joints, allowing more precise and intuitive control.
4. Targeted muscle reinnervation and regenerative peripheral nerve interfaces. These have been previously covered in the surgical section of this chapter; however, it is important to reiterate that a major barrier in more capable prosthetic limbs is the ability to control them. More advanced functions generally

require more inputs from the user. TMR and RPNI allow this increase in input channels from the user and help enable complex motions (Fig. 10.11). Other options to improve user input bandwidth include implantable electrodes in patient muscles or around peripheral nerves, brain-computer interfaces, and the use of machine learning and artificial intelligence to provide some levels of autonomous function in the hands.

5. Osseointegration. Currently approved by the FDA for trans-femoral level amputations, OI or bone-mounted attachment points for prostheses have been trialed with transhumeral (Fig. 10.12) and transradial levels, and even with thumb amputations. OI-mounted prostheses could be useful in upper extremities with short residual limbs or challenges with suspension systems.
6. Advances in residual limb and phantom limb pain. There are many groups studying the use of ketamine in acute and chronic amputation pain, peripheral nerve catheters to deliver continuous pain-blocking medication, cryoablation to denature peripheral nerves contributing to pain, and even high-concentration capsaicin patches. Severe phantom limb pain is notoriously difficult to treat, and it is important to explore all



• **Fig. 10.11** Target muscle reinnervation to improve myoelectric prosthetic control.



• **Fig. 10.12** Transhumeral osseointegration. Note the abutment, or prosthetic mount, extending percutaneously through the stoma, or hole, in the skin.

avenues of pain control to provide the highest quality of life possible.

Pediatric Amputation and Congenital Limb Difference

There are special considerations for amputation care in the pediatric patient with upper extremity amputation and congenital limb loss or limb difference. About 4 of every 10,000 babies each year are born with at least partial limb loss. In approximately 58% of cases, this is due to congenital deficiencies in which the limb bud fails to fully form in utero, but usually it does not have genetic or hereditary implications for the patient and the family. Teratogens and amniotic band syndrome are two of the most common causes. The International Society for Prosthetics and Orthotics terminology for congenital limb reductions is described in Box 10.7, which divides limb malformations into transverse or longitudinal. By definition, a child who has a transverse deficiency has no distal remaining parts beyond the named bone. For example, a child with a transverse radial deficiency has a normal upper arm and a portion of the radius but is missing the hand and fingers. Longitudinal deficiencies have distal portions present with a partial or total absence of a specific bone. This naming convention is sometimes used alongside the older Greek-based nomenclature shown in Box 10.8, which is still used in some environments. The most common congenital limb deficiency in the upper limb is a longitudinal partial or complete lack of the radius. Longitudinal hand reductions represent half of all congenital upper limb reductions, and multiple limb reductions are found in less than 20% of live births. Acquired amputations in children have a wide variety of causes but are most commonly due to trauma.³ Medical causes such as cancer, frostbite, vascular occlusion, or tumors also contribute.

Younger children with limb differences do not always experience the same sense of loss and psychological adjustment as an older amputee. In fact, it is quite common for children with

• BOX 10.7 ISPO Classification System for Congenital Upper Limb Reductions*

- *Transverse deficiency*: No remaining distal portions. Transverse level is named after the segment beyond which there is no skeletal portion.
- *Longitudinal deficiency*: Some remaining distal portions. Longitudinal deficiencies name the bones that are affected.

ISPO, International Society for Prosthetics and Orthotics.

*Any bone not named is present and of normal form.

• BOX 10.8 Original Classification Scheme for Congenital Upper Limb Reductions

- *Amelia*: Absence of a limb
- *Meromelia*: Partial absence of a limb
- *Phocomelia*: Flipper-like appendage attached to the trunk
- *Adactyly*: Absent metacarpal or metatarsal
- *Hemimelia*: Absence of half a limb
- *Acheiria*: Missing hand or foot
- *Aphalangia*: Absent finger or toe

congenital limb differences to not adopt a prosthetic at all. Frequently, they instead learn to function with their unique limb effectively. A prosthetic must be useful for a child to adopt, otherwise it may be discarded. Acceptance of the prosthetic by the parents is also important. The child's level of development and attention span also affect training sessions and prosthetic planning. Greater wear time of the prosthetic device has a positive correlation with comfort and with parental satisfaction.¹⁷

Prostheses can be fit once patients are ready to utilize them in their developmental stage. Young children may receive their first functional prosthesis once they are able to sit up and reach across midline to manipulate objects, which is generally around 6 months of age. This first prosthetic is usually a passive upper extremity device. Children may progress to more complex prosthetic components, such as an active terminal device, around 1 year of age, and myoelectric devices around 2 years of age.⁷ As children grow, their prosthetic requires frequent adjustments and replacements. Generally, a prosthesis or socket is replaced yearly in the first 5 years of life, then every 18 months between ages 5 and 12, and then biannually until the age of 21 years. Multilayered sockets are sometimes used so that each layer can be removed to accommodate growth. Length should also be adjusted to match patients' body proportions. Componentry such as single- and two-site control of myoelectric devices can be advanced as children age, with therapy repeated to allow for continued training and progress¹⁸ (Table 10.3).

In the pediatric amputee, different levels of amputation may be chosen during surgery to preserve skeletal growth potential, such as joint disarticulations. In cases where this is not possible, multiple surgeries may be needed over the course of their development. There are also children with congenital limb deficiencies who are surgically converted to an amputation to improve their function.

In skeletally immature children with an amputation of any extremity, one of the most common complications is terminal bony overgrowth of the residual limb. This occurs in 40% of children with amputations before the age of 12 years.⁴⁰ This may present as swelling, warmth, pain, or poor socket fit, and can be examined by x-ray, ultrasound, or other imaging. Surgical revision may be required if this overgrowth causes skin ulcerations or pain, limits range of motion, or limits prosthetic use. Recurrence rates remain high even after surgery.³⁹ After amputation, limb shaping and edema control are treated similarly to adults. Shrinker socks are not generally available in pediatric sizes, so limb edema is usually controlled with limb bandaging as an alternative prior to casting their prosthetic.²⁷ Children who use a prosthetic should be followed up at least once a year (or even more frequently during a growth spurt) to monitor their progress and growth and to troubleshoot issues with the device, supplies, equipment, skin, musculoskeletal complications, or functional deficits. For chil-

dren with an upper limb deficiency that does not use a prosthetic, every 2 to 3 years may be sufficient.

Summary

Upper limb amputation is a life-altering event that profoundly impacts physical function, emotional well-being, and social identity. While the condition carries significant psychological and functional challenges, advances in surgical technique, prosthetic technology, and interdisciplinary rehabilitation teams have greatly improved the outlook for affected individuals. There is a wide spectrum of upper limb amputations and therefore functional implications. Pre- and post-surgical planning plays a critical role in optimizing outcomes, including decisions about limb length, tissue management techniques, and pain management strategies such as Targeted Muscle Reinnervation (TMR) and Regenerative Peripheral Nerve Interface (RPNI). The rehabilitation process, ideally led by an experienced physiatrist, includes early interventions to manage pain, promote healing, and maintain joint mobility, followed by prosthetic evaluation, fitting, and then training phases tailored to individual goals and functional needs. There are many prosthetic types—passive, body-powered, myoelectric, and hybrid—and components, allowing patient-centric customization. It is important to be aware of potentially complex considerations such as bilateral amputations, pediatric limb differences, cognitive impairment, and comorbidities like skin issues, heterotopic ossification, and phantom limb pain, as these can easily and significantly change functional outcomes. Long-term follow-up is essential for monitoring prosthetic fit, addressing complications, and supporting psychological reintegration and vocational needs. Emerging technologies such as osseointegration, brain-computer interfaces, and 3D-printed limbs offer promising future advancements. Ultimately, successful rehabilitation hinges on an experienced interdisciplinary team, patient-centric goals, and ongoing support.

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TABLE 10.3 Age-Related Milestones for Pediatric Limb Loss

Upper Limb Prosthetic	Suggested Age
Passive terminal device	6 months when sitting independently
Active terminal device	12–15 months when ambulatory
Body-powered prosthesis	Around 1 year old
Myoelectric prosthesis	Around 2 years old

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